MB MIDI THRU 7

USER GUIDE

Firmware v3.0 • BLE + DIN MIDI merge & 7-way THRU router

Thank you for using the Mindburner MB MIDI THRU 7. This guide covers everything you need to get connected, configure the device from a browser, and get the best performance from it in your setup.

WHAT'S NEW IN v3.0

Channel Routing is now a full 16×16 matrix rather than a simple one-to-one map. Any input channel can be ticked to feed **any number** of output channels at once — so as well as reassigning a channel, you can now **clone** it across several outputs simultaneously. See Section 4 for details.

WHAT DOES IT DO?

The MB MIDI THRU 7 takes MIDI in from two sources — a standard 5-pin DIN cable and Bluetooth LE MIDI simultaneously — merges them, filters and routes them as you choose, and sends the result out through seven 5-pin DIN THRU ports. A built-in WiFi hotspot lets you configure everything from a phone or laptop browser, with a live MIDI monitor so you can see exactly what is passing through.

CONNECTING IT UP

Power — connect a USB cable to power the device (5V DC via USB).

DIN MIDI input — connect a standard 5-pin DIN MIDI cable from your sequencer, keyboard, DAW interface, or any MIDI OUT socket to the MIDI IN socket on the device. This is the default input and works the moment you power on — no setup needed.

Bluetooth LE MIDI input — the device advertises itself as **MB_MIDI_Thru** and accepts a BLE MIDI connection from any compatible device — a CME WIDI BUD plugged into a DIN port, a phone running a MIDI app, or a DAW with BLE MIDI support. Both DIN and BLE inputs are active at the same time and merged together.

THRU outputs — connect your synthesisers, drum machines, or other MIDI devices to any of the seven 5-pin DIN THRU output sockets. All seven carry the same merged, filtered, routed signal.

> **Tip:** The device only passes MIDI through — it never generates any of its own. It is completely safe to leave connected at all times.

1 WIFI MONITOR & CONFIG

Set up filtering, routing and live monitoring from your browser

The device hosts its own WiFi hotspot so you can configure it from any phone, tablet, or laptop — no app required, just a browser.

CONNECTING

Step 1. On your phone or laptop, open WiFi settings and connect to the network named **MB THRU7 CONFIG**. The password is **midi1234**.

Step 2. Open a browser and go to **192.168.4.1**. The configuration page will load.

Step 3. Make your changes and press **SAVE SETTINGS** at the bottom of the page. Settings are written to the device immediately and survive power-off.

> **Tip:** WiFi and BLE MIDI share the same radio on the ESP32. Both work simultaneously, but turning WiFi off when you don't need the monitor gives the best possible MIDI timing. See Section 2.

WHAT YOU CAN CONFIGURE

The page is divided into five sections:

Channel Activity	A live 16-block grid at the top of the page that lights up green whenever traffic arrives on a given MIDI channel. Useful for confirming at a glance which channels are active without reading the log.
Filters	Block entire message types from passing through — Notes, all CCs, Program Change, Pitch Bend, MIDI Clock, and Active Sensing can each be independently blocked.
Transpose	Shift all Note On and Note Off messages up or down by a fixed number of semitones (–24 to +24). Set to 0 to disable. The setting persists across power cycles.
CC Blocklist	Block individual CC numbers rather than all CCs at once. Common controllers are listed by name, plus a custom CC number field for any CC from 0 to 127.
Channel Routing	A 16×16 tick-matrix. Route any input channel to any output channel — or tick several boxes on a row to **clone** that channel across multiple outputs at once. See Section 4.

THE MIDI MONITOR

The lower half of the page shows a live scrolling log of every MIDI message passing through the device. Each line shows the source (DIN or BLE), the channel, the message type, the values, and — if Channel Routing sends it anywhere other than straight through — the actual output channel(s) it was sent to.

Pause / Resume — freeze the display while data keeps flowing, then resume to catch up. **Clear Monitor** — wipe the current log. **Auto Scroll** — keeps the newest message visible at the bottom. **Type filter** — show only Notes, CCs, Program Changes, Pitch Bend, Clock, or SysEx. **Pass filter** — show only passed or only filtered messages. **Hide Clock** — suppresses MIDI Clock messages, since they arrive 24 times per beat and can flood the display.

COLOUR CODING

Green	Note On / Note Off
Cyan	CC (controller)
Magenta	Program Change
Orange	Pitch Bend
Grey	Real-time (Clock, Active Sensing)
Pink	SysEx
Red	Filtered message (blocked, not passed)

2 TURNING WIFI ON AND OFF

Save power and improve timing when you don't need the monitor

WiFi is on by default. When you don't need the configuration page or live monitor, turning it off reduces the load on the processor and gives the best possible MIDI timing — particularly useful when passing MIDI Clock through, or when Channel Routing is cloning busy channels to several outputs.

METHOD 1 — WEB PAGE BUTTON (EASIEST)

At the top of the configuration page there is a banner showing the current WiFi state and a Turn WiFi OFF (or Turn WiFi ON) button. Pressing it saves the new state, shows a confirmation message, then restarts the device. The LED flashes 5 times to confirm the change, followed by the state indicator (see LED table below).

METHOD 2 — DOUBLE POWER-CYCLE (NO BROWSER NEEDED)

If you can't get to the browser — for instance, WiFi is already off — you can toggle it by power-cycling the device twice within 5 seconds:

Step 1. Power the device on. The LED rapid-flashes for about 2 seconds to show it is in the listening window.
Step 2. Power the device off within those 5 seconds.
Step 3. Power it back on. The LED flashes 5 times rapidly to confirm the toggle, then shows the new state.

If you leave the device running for more than 5 seconds after boot, the listening window closes and a normal power-off/on will not trigger a toggle — this prevents accidental toggling during normal use.

> **Tip:** The double power-cycle flag is stored in flash memory, not RAM, so it survives a real power cut. You can rely on it even if the device has been unplugged for a long time.

LED INDICATORS

LED	Meaning
3 quick flashes	WiFi is ON (shown on every boot)
1 long flash	WiFi is OFF (shown on every boot)
10 rapid flashes	Listening window open (power off now to prepare a toggle)
5 rapid flashes	WiFi state just toggled
Brief flash	MIDI message passed through
Double flash	MIDI message was filtered (blocked)

3 FILTERS

Block unwanted message types before they reach your gear

Filters let you stop specific types of MIDI message from passing through to the outputs. A filtered message is shown in red in the monitor but is not sent to any output port — filtering is applied once, before Channel Routing, so a blocked message is never cloned either.

AVAILABLE FILTERS

Notes	Blocks all Note On and Note Off messages on every channel.
All CCs	Blocks every CC message. Use the CC Blocklist instead if you only want to block specific controllers.
Program Change	Blocks Program Change messages on every channel.
Pitch Bend	Blocks Pitch Bend messages on every channel.
MIDI Clock	Blocks the 0xF8 Timing Clock real-time message. Useful if your downstream devices should only follow their own internal clock.
Active Sensing	Blocks the 0xFE Active Sensing keep-alive message. Some older devices behave unexpectedly if they receive Active Sensing and then it stops; blocking it here prevents that.

CC BLOCKLIST

Rather than blocking all CCs at once, the CC Blocklist lets you block individual controller numbers. Common ones are listed by name:

Mod Wheel	CC 1 — modulation wheel
CC 2	CC 2 — breath controller
Volume	CC 7 — channel volume
Pan	CC 10 — stereo pan
Expression	CC 11 — expression pedal
Sustain	CC 64 — sustain pedal
CC 65	CC 65 — portamento on/off
CC 66	CC 66 — sostenuto
CC 67	CC 67 — soft pedal
CC 68	CC 68 — legato
CC 91	CC 91 — reverb send
CC 92	CC 92 — tremolo depth
CC 93	CC 93 — chorus send
CC 94	CC 94 — detune depth
Custom CC	Enter any CC number from 0 to 127 to block it.

Tip: Filters and the CC Blocklist work on both DIN and BLE input simultaneously.

4 CHANNEL ROUTING

Route — or clone — any input channel to any output channel(s)

Channel Routing lets you change which output channel a message comes out on, regardless of which channel it arrived on. As of v3.0 this is a full 16×16 **matrix** rather than a simple one-to-one map, so an input channel is no longer limited to a single destination — tick more than one output for a channel and it is **cloned** onto all of them simultaneously.

HOW IT WORKS

The routing page shows a 16×16 grid: one row per input channel, one column per output channel. Tick the boxes on a row for the output channel(s) you want that input to feed. By default each row has just its own diagonal box ticked (CH1 in → CH1 out, CH2 in → CH2 out, and so on).

To reroute, untick the diagonal box and tick a different column on that row, then press **SAVE SETTINGS**. To clone, simply tick additional boxes on the same row — every ticked output receives an identical copy of everything arriving on that input channel.

If every box on a row is unticked, that input channel is effectively muted — its messages are received but sent nowhere. The monitor marks these as **[NO OUTPUT]** so it's obvious at a glance if a row has been left empty by mistake.

EXAMPLE USES

Fixing a channel mismatch — Your sequencer sends on CH1 but your synth is set to receive on CH3. Untick CH1→CH1 and tick CH1→CH3 instead.

Cloning a channel to layer two synths — Both synths should play the same part but are set to receive on different channels — say CH1 and CH5. Tick both CH1→CH1 and CH1→CH5 on the CH1 row; every note, CC and bend sent on CH1 now reaches both synths simultaneously, no re-patching required.

Sending one clock/transport channel to several devices on different channels — If your downstream gear is fixed to different receive channels but you only have one controller channel to send from, tick that input row across every output channel those devices listen on.

Isolating a device — Untick every row except the one feeding CH16, so only a device listening on CH16 receives anything.

Layering two synths on the same channel from two different controller channels — Both synths listen on CH1. Tick CH1→CH1 for the first synth's part, and also tick CH2→CH1 so a second controller channel merges onto the same output.

ORDER OF OPERATIONS

Filters and the CC Blocklist are applied first — a blocked message is dropped before Channel Routing ever sees it, so it is never cloned. Transpose is then applied once to the note value, and finally the (possibly transposed) message is sent out on every ticked output channel for that row. All clones of a message carry the same transposed note number.

> **Tip:** The MIDI Monitor log shows the real output channel(s) a message was sent to whenever it differs from a plain 1:1 pass-through — useful for confirming a clone is actually reaching every output you expect.

> **Tip:** Cloning a busy channel (dense Notes/CC) across many outputs multiplies the bytes sent out the single DIN THRU UART. This is fine for normal use, but if you clone several very busy channels across many outputs simultaneously you can approach the MIDI baud-rate ceiling, which shows up as added latency or jitter rather than dropped notes.

5 TRANSPOSE

Shift notes up or down by semitones

The Transpose setting shifts every Note On and Note Off message by a fixed number of semitones before it reaches the output ports. It works on both DIN and BLE input simultaneously, is applied once per incoming message (so every cloned copy from Channel Routing carries the same transposed note), and the setting is saved to flash so it persists across power cycles.

SETTING IT UP

Step 1. Connect to the web page (see Section 1) and scroll to the Transpose card.

Step 2. Enter a value between –24 and +24 semitones. Positive values shift notes up; negative values shift down. 0 disables transpose entirely (the default).

Step 3. Press **SAVE SETTINGS**. Transpose takes effect immediately — no reboot needed.

> **Tip:** Transpose is applied to Note On and Note Off only — Pitch Bend, CC, and other messages pass through unchanged. If a transposed note number would go below 0 or above 127 it is clamped to the nearest valid value rather than wrapping around.

EXAMPLE USES

Playing a bass synth from a melody keyboard — Your keyboard plays in the 4th octave but your bass synth works best an octave lower. Set Transpose to –12 and the synth drops a full octave without touching the keyboard.

Matching two synths tuned to different octaves — One synth always sounds an octave higher than expected. Route that synth onto its own channel and set Transpose to –12, and the two now match.

Capo equivalent for a MIDI guitar controller — Need to play in a different key without retuning? Transpose shifts every note by the equivalent number of semitones.

6 MIDI TIMING NOTES

What to know when mixing DIN and Bluetooth sources

BLE MIDI LATENCY

DIN MIDI is a direct serial connection — a Note On arrives in about 1ms. BLE MIDI batches messages into packets on a connection interval, typically 7.5–15ms depending on the host device. Any BLE message already carries this delay before it reaches the device. For playing notes and sending controllers this is generally imperceptible, but for tight timing work DIN is always the more reliable choice.

ONLY USE ONE CLOCK SOURCE AT A TIME

If your DIN source is sending MIDI Clock and a BLE device is also sending Clock, both will arrive at the outputs and your downstream devices will receive a doubled, corrupted clock signal. Always use the Clock filter to block Clock from whichever source you are not using as master.

> **Tip:** A reliable split: use DIN for clock and transport, BLE for notes and controllers. Filter Clock on the BLE input and you get the best of both without any interference.

WIFI AND TIMING

When WiFi is on and a browser is actively polling the monitor page, the device occasionally takes a few extra milliseconds to read incoming DIN bytes. The bytes are not lost (the hardware UART buffers them), but they arrive at the outputs slightly late, which shows up as jitter on Clock and other timing-sensitive messages.

For best timing during performance or recording, turn WiFi off. BLE MIDI continues to work regardless of WiFi state.

BLE AND DIN ARRIVING SIMULTANEOUSLY

In normal use it is very unlikely that a DIN byte and a BLE message arrive at exactly the same instant, but if they do the device handles them sequentially. For most real-world playing situations this is completely transparent.

CLONED CHANNELS AND THROUGHPUT

Channel Routing can send one input channel's data out of several output channels at once (Section 4). Each clone is a full extra message on the single DIN THRU UART, so cloning dense traffic (fast-moving CCs, chords) across many outputs increases total bytes sent. This is not a concern for normal playing, but with several very busy channels cloned widely it's possible to approach the 31,250 baud ceiling, which shows up as added latency rather than dropped notes.

7 FIRMWARE UPDATES

Keep your device up to date over USB

From time to time new firmware becomes available, adding features or fixing issues. Updates are free and user-installable — no soldering, no special software, and no need to send the device back.

WHAT YOU NEED

A USB cable — connect the device to your PC.

A browser — Google Chrome, Microsoft Edge, or Opera. Safari and Firefox do not support the WebSerial API used by the installer.

The right driver — Mindburner ESP32 devices use a Silicon Labs CP2102 chip. If your PC does not recognise the device when you plug it in, download and install the CP210x driver from the Silicon Labs website. A step-by-step guide is included in the driver download folder available from mindburner.co.uk.

INSTALLING AN UPDATE

Step 1. Visit the firmware page at mindburner.co.uk and find the MB MIDI THRU 7 entry.

Step 2. Connect the device with a USB cable.

Step 3. Click CONNECT on the firmware page. A small popup lists available serial ports — select the one corresponding to your device (usually named CP210x or similar).

Step 4. Click INSTALL and wait for the progress bar to complete. The device restarts automatically when done.

> **Tip:** Your saved settings (filters, routing matrix, WiFi state) are normally preserved across a firmware update. If you are ever asked whether to fully erase the device first, leave that option off to keep your settings. If they do get reset, it only takes a moment to set them again from the web page.

TECH SPECIFICATIONS

MIDI Input	5-pin DIN + Bluetooth LE MIDI (simultaneously)
MIDI Output	Seven 5-pin DIN THRU ports
Channel Routing	16×16 matrix — route or clone any input channel to any number of output channels
Bluetooth	BLE MIDI — advertises as MB_MIDI_Thru
WiFi Config	Built-in AP: MB THRU7 CONFIG / midi1234 / 192.168.4.1
Power	5V DC via USB
Firmware	Free updates over USB from any browser

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